

Job Market Analysis (SQL & Power BI) - Dashboard Documentation

This dashboard is designed to help users understand the current job market for Data Analysts. The analysis focuses on salary trends, skill demand, remote work opportunities, and the skills that offer the best balance between market demand and earning potential. SQL is used for data extraction and analysis, while Power BI is used to create interactive visualizations and insights.

Page 1: Executive Overview

The Executive Overview page provides a high-level summary of the job market. Users can quickly understand the size of the market, salary trends, and hiring activity. Key performance indicators include the total number of job postings, total hiring companies, percentage of remote jobs, average salary, highest salary offered, and the number of unique skills mentioned in job postings. Recommended visualizations include salary distribution charts, remote versus non-remote job comparisons, maps showing job locations, and charts highlighting the companies with the most openings.

Page 2: Top Paying Jobs

This page focuses on identifying the highest-paying Data Analyst roles. It helps job seekers understand which positions offer the greatest earning potential and which organizations provide competitive compensation. Important metrics include the highest-paying role, the average salary among top positions, and the number of companies represented in the top salary category. Visualizations can include bar charts ranking the top-paying jobs, company salary comparisons, and detailed salary tables.

Page 3: Most In-Demand Skills

The objective of this page is to identify the skills most frequently requested by employers. Understanding demand helps professionals prioritize learning and career development efforts. Commonly requested skills often include SQL, Excel, Python, Tableau, and Power BI. Visualizations such as bar charts, treemaps, and skill frequency analysis can help users quickly identify market trends.

Page 4: Highest Paying Skills

While some skills are popular, others command significantly higher salaries. This page highlights the technologies and tools associated with premium compensation. Skills such as Azure, AWS, BigQuery, Python, Tableau, and SQL are often linked to higher-paying positions. Charts comparing average salaries by skill can help users understand which competencies generate the greatest financial value.

Page 5: Optimal Skills Analysis

This page combines salary information and market demand to identify the most valuable skills for career growth. The goal is to help professionals invest their learning time wisely. A demand-versus-salary analysis can classify skills into four categories: High Demand and High Salary, High Demand and Lower Salary, Low Demand and High Salary, and Low Demand and Low Salary. Skills that fall into the High Demand and High Salary category are generally the best choices for career advancement.

Advanced KPI Suggestions

Additional metrics can improve the analytical value of the dashboard. Examples include median salary, salary growth potential, top employer salary, top skill salary, average salary by location, and demand-to-salary ratios. These metrics provide deeper insights into market behavior.

Interactive Features

To improve user experience, the dashboard should include filters for job title, company, location, skill, and remote work availability. Dynamic titles, drill-through functionality, custom tooltips, and navigation buttons can make the dashboard more interactive and professional.

Business Impact

This project demonstrates practical business intelligence skills by transforming raw job posting data into actionable insights. It showcases SQL querying, data modeling, dashboard development, KPI creation, and analytical storytelling. The final dashboard can help students, job seekers, and professionals make informed career decisions based on real market data.